Lecture 31 (April 18), continued

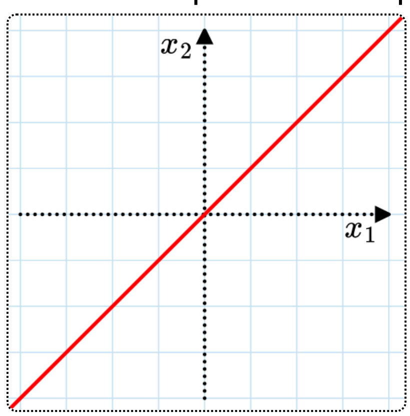
The sequence of numbers $\{n^{n-2}\}$ counts a number of interesting objects, related to 'hyperplane arrangements', for which the next few lectures will be an introduction.

- the number of spanning trees of K_n, $t(K,n) = n^{n-2}$.
- the number of parking functions of size n, $(n+1)^{n-1}$.
- the number of regions of Shi arrangements, $(n+1)^{n-1}$.
- $\mathrm{Vol}(\text{permutahedron } \Pi_n) = n^{n-2}$.

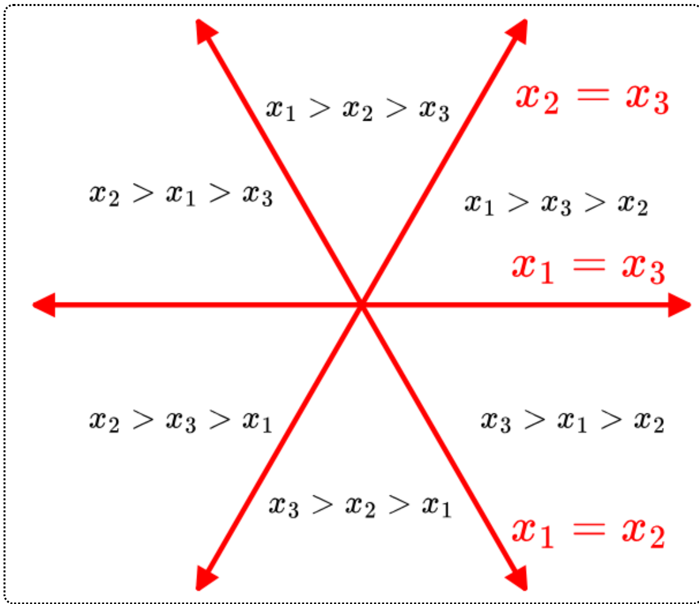
Braid arrangements (related to the braid group)

The braid arrangement Br_n in $\mathbb{R}^n = (x_1, x_2, \ldots, x_n)$ consists of the hyperplanes of the form $x_i - x_j = 0$ for all $i \neq j$.

For example, Br_2 is just one line that separates the plane into two regions.



$n = 3$ comprises six planes all perpendicular to $x_1 + x_2 + x_3 = 0$, so here is a cross-section (or projection onto that plane):

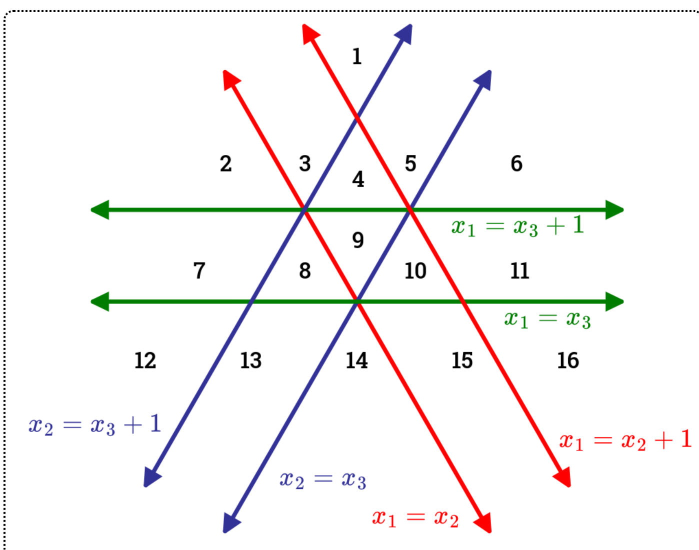


Note that it splits $\mathbb{R}^3$ into 6 regions based on the ordering of x_1, x_2, x_3: $x_1 > x_2 > x_3$, $x_1 > x_3 > x_2$, and so on. As a result, in general, Br_n creates $n!$ regions.

Shi arrangements

These are very similar to braid arrangements but we 'duplicate' each hyperplane. $\mathrm{Shi}_n \in \mathbb{R}^n$ includes $x_i - x_j = 0$ and $x_i - x_j = 1$ for all $i < j$.

For instance, for $n = 3$, there are 16 regions, which is indeed $(n+1)^{n-1}$ (again looking at the cross-section with $x_1 + x_2 + x_3 = 0$):



Lecture 32 (April 23)

Parking functions

We want to consider a problem where there are n cars that drive on a one-way road towards n parking spaces. We have a **preference function** $f\colon [n] \to [n]$ which can also be written as $f = (f_1, \cdots, f_n)$, meaning that car i prefers to park in spot f_i.

The procedure is as follows:

- In order, each car i drives toward spot f_i.
- If the spot is available, they will park there
- Otherwise, the car will continue moving forward and park in the next available spot.
  We call a preference function a **parking function** if all n cars can successfully park.

For example, if $n = 4$ and $f = (3, 3, 1, 1)$, the cars can park in spots 3, 4, 1, and 2, respectively. However, if $f = (3, 3, 2, 2)$, car 4 will not be able to find a parking spot. So, only the first f is a parking function.

Lemma 1 (Conditions for parking functions).

Given $f = (f_1, \cdots, f_n)$, the following are equivalent:

(1) f is a parking function.
(2a) For any $k \in [n]$, there are at most k entries f_i that are at least $n - k + 1$.
(2b) For any $k \in [n]$, there are at least k entries f_i that are at most k.
(3) There exists a permutation $w_1, \cdots, w_n$ of $[n]$ such that $f_i \leq w_i$ for all i.

(2a) and (2b) are equivalent. The idea is to show that (1) implies (2a) implies (3) implies (1). One corollary of this is that any permutation of a parking function is also a parking function.

For example, for $n = 3$, we have six permutations of $(1, 2, 3)$, three permutations each of $(1, 2, 2)$, $(1, 1, 3)$, and $(1, 1, 2)$, and $(1, 1, 1)$, for a total of

$6+3+3+3+1=4^2$ parking functions. We have the following theorem:

Theorem 2 (Kreweras (1980)).

There are $(n+1)^{n-1}$ parking functions of size n.

Proof. Consider a slightly different problem where there are $n+1$ parking spots on a circular road. Now our preference functions are $f\colon [n] \to [n+1]$, and with the same procedure for parking, it's clear that every one will yield a successful parking arrangement (since any car can keep going around the road). There are a total of $(n+1)^n$ of these.

For each function, there is one open parking spot at the end. Let F_i be the number of preference functions such that spot i is left open. By symmetry, each of the F_is are equal (e.g. shift all the numbers in any preference function modulo $n+1$), so $F_{n+1}=(n+1)^{n-1}$. Note that these arrangements where $n+1$ is open at the end are exactly the same that are valid parking functions in our original problem (since no car would have to go past spot n), so we are done.

Theorem 3 (Weakly increasing parking functions).

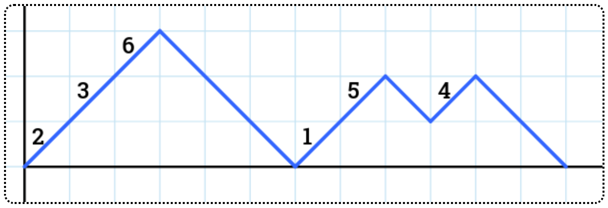
The number of weakly increasing parking functions, where $f_1 \leq f_2 \leq \cdots \leq f_n$, is the Catalan number C_n.

This follows from (2b) of Lemma 1, since it implies $f_i \leq i$ for all i, and it's not difficult to biject these sequences to Dyck paths.

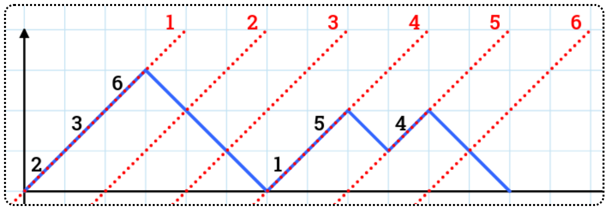
Correspondence with labeled trees

Since $(n+1)^{n-1}$ is also the value that appears in Cayley's Theorem, we want to biject parking functions to labeled trees on $n+1$ vertices. The idea is to first biject them with "labeled Dyck paths with $2n$ steps".

To create a labeled Dyck path, we start with a standard Dyck path and label all of the up steps with $[n]$ so that the labels of consecutive up steps increase. For example, the below Dyck path has $2<3<6$ and $1<5$.



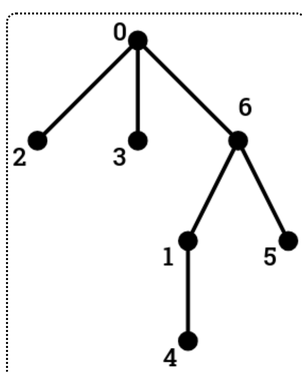
We biject this Dyck path with a parking function by drawing diagonals and letting f_i be the diagonal that the label i appears on. The above corresponds to $(4, 1, 1, 5, 4, 1)$, since e.g. the label 1 is on diagonal 4:



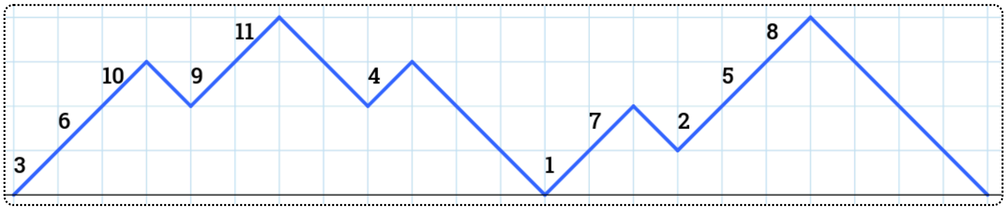
It is not hard to construct a lattice path from any parking function, and the condition of the parking function being valid is equivalent to the path staying above the x-axis.

To map them to labeled trees, we once again go diagonal-by-diagonal, adding children to the leftmost vertex in the tree not yet considered.

1. $2, 3, 6$ are on the first diagonal, so we connect them to 0. They could still have children, so we let them correspond to the next few diagonals.
2. No numbers are on the next two diagonals, so we do not add children to 2 or 3 (the two leftmost 'open' vertices).
3. Now only 6 is open. The next diagonal has 1 and 5, so we make them children of 6.
4. The next diagonal has 4, so we connect it to the leftmost open vertex, 1. The result is the following.

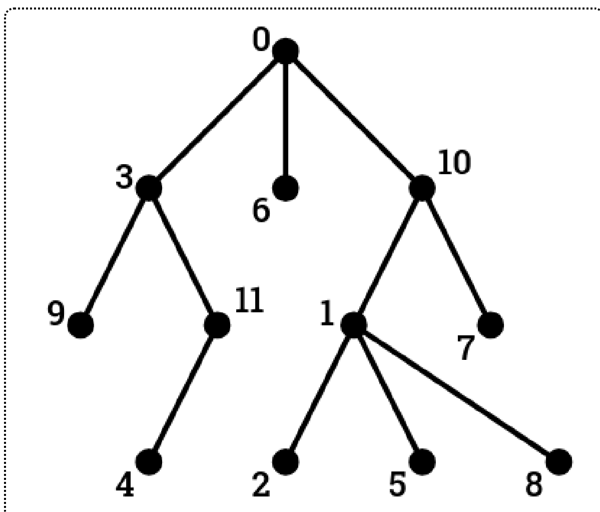


For another example:



- $3, 6, 10$ are children of 0 (open vertices: $3, 6, 10$)
- $9, 11$ are children of 3 (open vertices: $9, 11, 6, 10$)
- nothing is a child of 9 (open vertices: $11, 6, 10$)
- 4 is a child of 11 (open vertices: $4, 6, 10$)
- nothing is a child of 4 or 6 (open vertices: 10)
- $1, 7$ are children of 10 (open vertices: $1, 7$)
- $2, 5, 8$ are children of 1 (open vertices: $2, 5, 8, 7$)
- there are no more vertices to add.

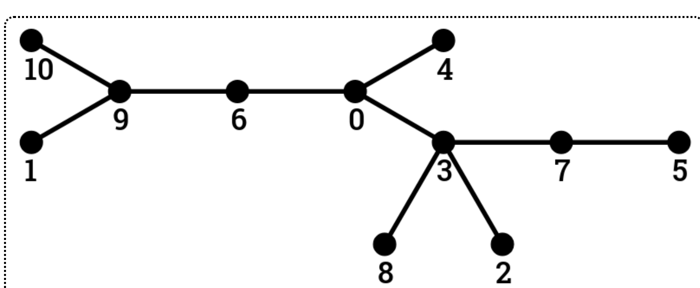
The result is the following.



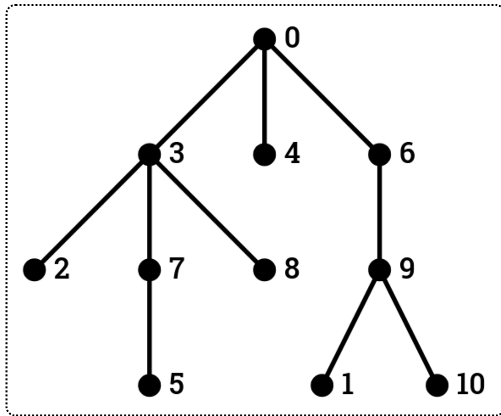
As an exercise, try to reverse this procedure.

Lecture 33 (April 25)

We'll recap the bijection from last class. Let's start with a tree this time (labeled with 0 through n):



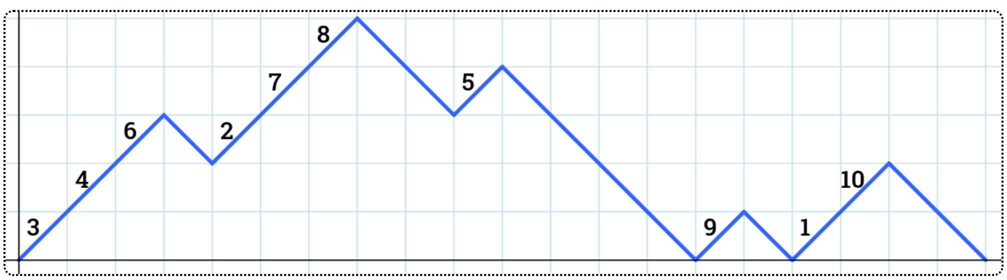
First, we rearrange it so that it's rooted at 0 and all children are arranged in increasing order:



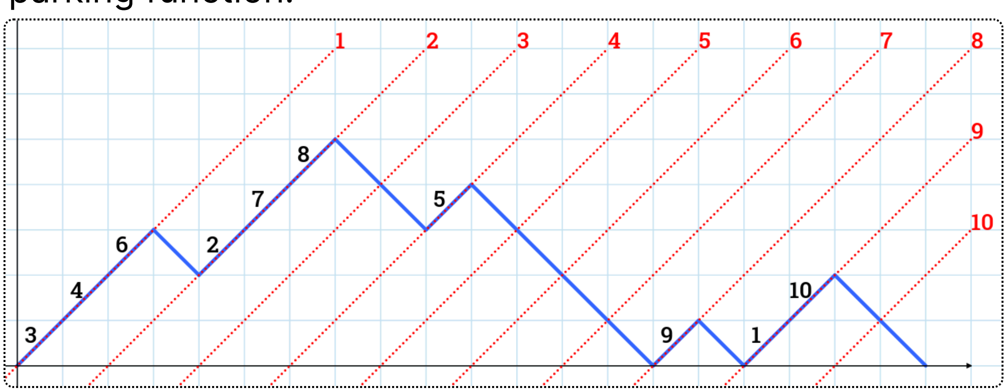
We consider the depth-first-search traversal order of the graph (i.e. keep going down the leftmost edge until you can't anymore, and backtrack until you get a new edge to follow), and keep track of each node's children along the way:

vertex	0	3	2	7	5	8	4	6	9	1	10
children	3, 4, 6	2, 7, 8		5				9	1, 10		

To construct our labeled Dyck path, we go through each column of the above table. We add a labeled up step for each child, and a down step afterwards (marking that we move to the next column):



Finally, we draw the diagonals and note which one each label appears in, to obtain our parking function.



$$\Longrightarrow f = \begin{pmatrix} 9 & 2 & 1 & 1 & 4 & 1 & 2 & 2 & 8 & 9 \end{pmatrix}$$
$$\begin{matrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 & 10 \end{matrix}$$

Statistics

Let's look at some statistics on parking functions. First consider the generating function p_n that tracks how much extra distance cars had to drive past their preferred spot:

$$p_n(x) = \sum_{f=(f_1,...,f_n)} x^{\binom{n+1}{2} - \sum_{i=1} f_i}$$

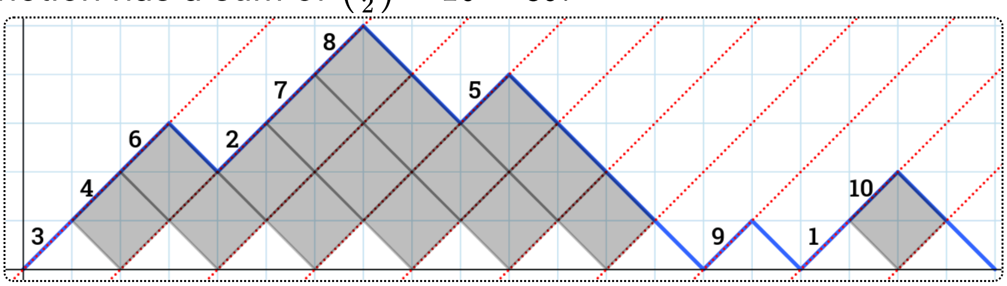
For example, if $n = 3$, this is $6 + 6x + 3x^2 + x^3$. Each permutation of $(1, 2, 3)$ contributes 1, each permutation of $(1, 2, 2)$ or $(1, 1, 3)$ contributes x, each permutation of $(1, 1, 2)$ contributes x^2, and $(1, 1, 1)$ contributes x^3.

Lemma 4 ($p_n(x)$).

$$p_n(x) = \sum x^{\text{area(P)}}$$

where the sum is over all labeled Dyck paths P with $2n$ steps, and $\text{area}(P)$ is the area of diamonds under the path of P. This directly results from our bijection.

For example, in the above example, $\text{area}(P)$ is 16, and indeed its associated parking function has a sum of $\binom{11}{2} - 16 = 39$.



Inversions in trees

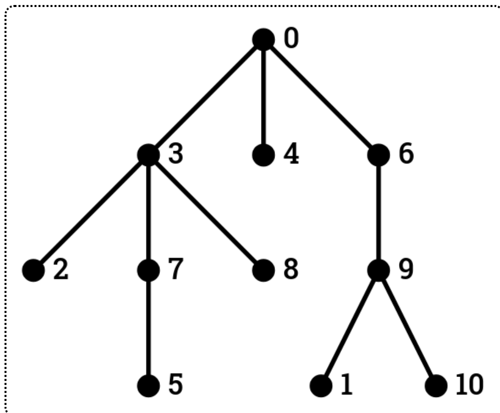
Since we biject to labeled trees, let's try to find a statistic for labeled trees that corresponds to $p_n(x)$. Suppose T is a labeled tree with $n + 1$ vertices (with labels in $0, 1, 2, \ldots, n$).

Definition (Labeled tree inversions)**.**

A pair (i, j) with $1 \le i < j \le n$ is an **inversion** in T if j is on the shortest path from the root 0 to i.

$\operatorname{inv}(T)$ denotes the number of inversions in T.

For example, in our example tree,



our inversions are $(2, 3)$, $(5, 7)$, $(1, 6)$, and $(1, 9)$. So, $\operatorname{inv}(T) = 4$.

Theorem 5 (Tree-inversion polynomial (Kreweras 1980))**.**

The sum $I_n(x) = \sum x^{\operatorname{inv}(T)}$ over all labeled trees on $n + 1$ vertices equals $p_n(x)$.

For example, we can verify that $I_2(x) = p_2(x) = 2 + x$.

Unfortunately, our bijection does not immediately prove this theorem (our example had $\operatorname{inv}(T) = 4$ and $\binom{n+1}{2} - \sum f_i = 16$).

Kreweras constructed a much more complicated bijection to prove this using induction, via the following recurrence relation:

$$
\begin{aligned}
I_0(x) &= 1 \\
I_n(x) &= \sum_{k=1}^{n} \binom{n-1}{k-1} (1 + x + x^2 + \cdots + x^{k-1}) I_{k-1}(x) I_{n-k}(x)
\end{aligned}
$$

This recurrence is similar to one you may have seen for the Catalan numbers (and it makes some sense, since Catalan numbers count unlabeled Dyck paths while parking functions are labeled Dyck paths).

$$C_0 = 1$$
$$C_n = \sum_{k-1}^{n} C_{k-1} C_{n-k}$$

All of these recurrences are proved by "decomposing" each Dyck path/tree/etc into two smaller instances of the same object. The idea for decomposing labeled tree is to "cut off" the subtree T_1 containing vertex 1 at the root. k represents the number of vertices in T_1. Selecting these vertices yields $\binom{n-1}{k-1}$ and the factor $(1 + x + x^2 + \ldots)$ comes from selecting the root of T_1.

This tree-inversion polynomial $I_n(x)$ has some interesting values:

$$I_n(1) = (n+1)^{n-1}$$
$$I_n(0) = n!$$
$$I_n(-1) = ?$$

n	0	1	2	3	4	5	6
$I_n(-1)$	1	1	1	2	5	16	61

We will study this sequence next week. (Look it up in OEIS)

Lecture 34 (April 28)

We will continue examining $p_n(x)$ and $I_n(x)$. Using our recurrence, we can evaluate the first few vallues.

$$p_0 = 1$$
$$p_1 = p_0 p_0 = 1$$
$$p_2 = p_0 p_1 + (1+x) p_1 p_0 = 1 + (1+x) = 2 + x$$
$$p_3 = p_0 p_2 + 2(1+x) p_1 p_1 + (1 + x + x^2) p_2 p_0 = 6 + 6x + 3x^2 + x^3$$
$$p_4 = p_0 p_3 + 3(1+x) p_1 p_2 + 3(1 + x + x^2) p_2 p_1 + (1 + x + x^2 + x^3) p_3 p_0 = \ldots$$
$$\ldots$$

One of the problems on the next problem set will be to prove this recurrence relation for parking functions p_n. There is a proof that I_n satisfies this recurrence in previous years' lecture notes.

Alternating permutations

Last class, we introduced the sequence $I_n(-1) = 1, 1, 1, 2, 5, 16, 61, \ldots$. This sequence has many different names and definitions, e.g. involving the following:

Definition (Alternating permutations).

An **alternating permutation** (or an up-down permutation, or a zigzag permutation, etc.) is a permutation w satisfying the following:

$$w_1 < w_2 > w_3 < w_4 > w_5 < \ldots$$

A_n is the number of alternating permutations in S_n.

André's Problem is to determine values of A_n. The numbers A_n are also called "zigzag numbers". These numbers have slightly different behavior depending on the parity of n: the odd ones are sometimes known as "tangent numbers" and the even numbers are "secant numbers" or "Euler numbers" (not to be confused with Eulerian numbers).

Theorem 6 (p_n and A_n).

$$p_n(-1) = A_n.$$

There are a few ways to prove this theorem, including induction using recurrence relations and the involution principle (pairing up most parking functions in a way that the parity of $\sum f_i$ changes).

Inductive proof. Let $\tilde{A}_n := p_n(-1)$. From our recurrence relation for p_n, we see that plugging in $x = -1$ causes half of the terms to cancel out and

$$\begin{aligned} \tilde{A}_n &= \sum_{\substack{k \geq 1 \\ k \text{ odd}}} \binom{n-1}{k-1} \tilde{A}_{k-1} \cdot \tilde{A}_{n-k} \\ \tilde{A}_0 &= 1. \end{aligned}$$

We want to show that A_n satisfies this recurrence as well. The base case checks out and we need to somehow split up an alternating permutation into two smaller ones, of size $k-1$ and $n-k$. The idea is to classify permutations based on where 1 appears.

Suppose $w_k = 1$; k must be odd since 1 is not more than any other entry in the permutation. We now have two alternating sequences, $w_1 < w_2 > w_3 < \cdots < w_{k-1}$ of length $k-1$ and $w_{k+1} > w_{k+2} < \ldots w_n$ of length $n-k$.

There are $\binom{n-1}{k-1}$ ways to pick which numbers appear on which side of 1, A_{k-1} ways to pick an alternating permutation of $\{w_1, \ldots, w_{k-1}\}$, and A_{n-k} ways to pick a down-up permutation of the remaining numbers for $(w_{k+1}, \ldots, w_n)$. Thus each value of k contributes $\binom{n-1}{k-1}\tilde{A}_{k-1} \cdot \tilde{A}_{n-k}$, as desired.

Lemma 7 (Another relation).

A_n also satisfies

$$A_n = \sum_{\substack{k \geq 2 \\ k \text{ even}}} \binom{n-1}{k-1} A_{k-1} A_{n-k}.$$

The proof is analogous (except it considers the position of n, not 1.)

Exponential generating functions

We introduce a new type of generating function.

Definition (EGFs).

The **exponential generating function** (EGF) for $\{A_n\}_{n\geq 0}$ is

$$A(x) = \sum_{n\geq 0} A_n \frac{x^n}{n!}.$$

Generally, ordinary generating functions are better suited for counting unlabeled objects and EGFs are better for labeled objects (like permutations).

We can also define $A^{\text{even}}(x)$ and $A^{\text{odd}}(x)$ which are the sums over even and odd n, respectively.

In our specific case with alternating permutations, we want to use our recurrences to determine a closed form for the EGF. Indeed, thanks to the $\binom{n-1}{k-1}$ term, we can rewrite them as e.g.

$$\begin{aligned}
\frac{A_n}{(n-1)!} &= \sum_{k \text{ even}} \left(\frac{A_{k-1}}{(k-1)!} \cdot \frac{A_{n-k}}{(n-k)!} \right) \\
\implies \frac{A_n}{(n-1)!} x^{n-1} &= \sum_{k \text{ even}} \left(\frac{A_{k-1}}{(k-1)!} x^{k-1} \cdot \frac{A_{n-k}}{(n-k)!} x^{n-k} \right) \\
\implies \sum_{n \geq 1} \frac{A_n}{(n-1)!} x^{n-1} &= \left(\sum_{k \text{ even}} \frac{A_{k-1}}{(k-1)!} x^{k-1} \right) \cdot \left(\sum_{\ell \geq 0} \frac{A_\ell}{\ell!} x^\ell \right) + 1 \\
\implies \frac{d}{dx} A(x) &= A^{\text{odd}}(x) \cdot A(x) + 1 \\
\implies \frac{d}{dx} A^{\text{even}}(x) &= A^{\text{odd}}(x) \cdot A^{\text{even}}(x) \\
\text{and } \frac{d}{dx} A^{\text{odd}}(x) &= A^{\text{odd}}(x) \cdot A^{\text{odd}}(x) + 1
\end{aligned}$$

(Note: ℓ is just $n-k$ renamed since we summed both sides over all n, and we added an extra 1 to account for the $A_0 = 1$ base case of the recurrence.)

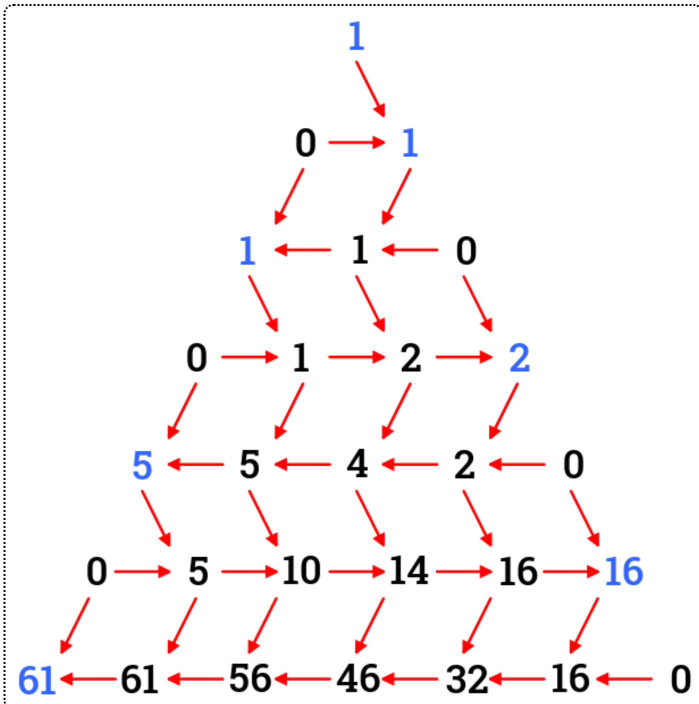
We have the initial conditions $A^{\text{even}}(0) = 1$ and $A^{\text{odd}}(0) = 0$. It turns out that the solution to these differential equations is very nice:

Theorem 8 (Secant and tangent)**.**

$$\begin{aligned}
A^{\text{even}}(x) &= \sec(x) = \frac{1}{\cos(x)} \\
A^{\text{odd}}(x) &= \tan(x) \\
\implies A(x) &= \sec(x) + \tan(x).
\end{aligned}$$

Seider-Entringer-Arnold triangle

Here is a triangle of numbers, also known as the "Euler-Bernoulli triangle". Each row consists of partial sums of the row above it, though you alternate between starting from the left and starting from the right.



Note how the values of A_n appear on either side of the triangle, with secant/Euler numbers on the left and tangent numbers on the right. As an exercise, try to prove that this does indeed happen.